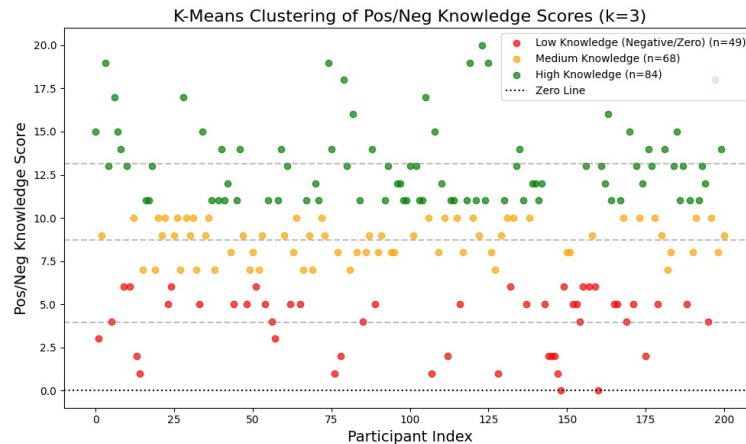


Detailed Analysis: K-Means Clustering on Pos/Neg Scores

A K-Means algorithm ($k = 3$) was deployed on the penalized Positive/Negative scores to objectively categorize the participants into three knowledge tiers.



PosNeg K-Means

Cluster Psychology & Medical Implications

With negative penalties applied, the K-Means algorithm naturally separated the cohort based on the severity of their misconceptions:

- 1. Low Knowledge (Red / Negative Tier):**
 - **Characteristics:** This cluster is dominated by participants who guessed incorrectly frequently or lacked basic knowledge entirely.
 - **Implication:** This group requires urgent, targeted medical education to unlearn dangerous myths (such as transmission vectors or false cures).
- 2. Medium Knowledge (Orange / Moderate Tier):**
 - **Characteristics:** Participants who understand the foundational basics of thalassemia (e.g., inherited, requires screening) and avoided guessing incorrectly on the complex medical questions.
- 3. High Knowledge (Green / Expert Tier):**
 - **Characteristics:** This elite cluster successfully identified complex medical complications (iron overload, endocrine issues) and possessed almost zero misconceptions. They achieved high positive scores without accumulating negative penalties.